

House Price Prediction

- **Dataset Description**

The Housing Prices Dataset (Kaggle) contains 545 rows and 13 columns. The target variable is price (house sale price in Indian Rupees). Features include area (sq ft), bedrooms, bathrooms, stories, parking, and binary amenity flags (mainroad access, guestroom, basement, hot water heating, air-conditioning, preferred area) plus furnishing status. No missing values were found; 0 duplicate rows were removed. Categorical columns were encoded: binary yes/no fields mapped to 1/0 and furnishing status one-hot encoded into three columns.

- **Model Performance Comparison**

Metric	Linear Regression	Random Forest
MAE	Rs 9,70,043	Rs 10,09,083
RMSE	Rs 13,24,507	Rs 13,91,456
R ² Score	0.6529	0.6170

Linear Regression performed slightly better on this dataset ($R^2 = 0.6529$ vs 0.6170), suggesting the feature-price relationships are largely linear.

- **Insights**

Q1. Which features influence house price the most?

ANS. According to given dataset, Area has the most influence in predicting the house price. Also, number of bathrooms, air-conditioning and location has a good influence on house price.

Q2. How accurate was the model?

ANS. In linear Regression, R^2 is 0.6529 i.e., there is a 65% of the variance in house prices whereas R^2 is 0.617 showing that the relationships in this dataset are largely linear.

Q3. What surprised you?

ANS. Dataset has no duplicate rows or missing values. Also, The guestroom and basement features has very weak correlation with price whereas air-conditioning and bathrooms carry far more weight with buyers in this dataset.

- **Recommendation for a real estate business**

Listings should prominently feature total area and the number of bathrooms as primary value drivers. Properties in preferred areas with air-conditioning consistently command premium prices i.e., agents should use these attributes to justify higher asking prices, prioritise such properties for marketing investment, and counsel sellers to invest in air-conditioning as a high-ROI upgrade before listing.

